

Backup and clean-build

Build folders and logs grow fast, waveforms, object files, run logs

Timestamp, exclude, then clean

- Name archives with a date stamp so versions do not overwrite each other
- Exclude build and logs from the backup when you only need sources, docs, and scripts
- For clean: list files under build and logs first
- Align exclude patterns with your ignore file so archives and Git stay consistent

Browser lab

Digital Design and Verification Platform

HomeToolsBackup / clean

Make a timestamped backup before cleaning – know what is safe to delete (build/, logs, waves) vs what is source.

Starter example: Project has source plus generated build/, logs/, and a root sin_vcd. Backup first, dry-run clean, then clean for real.

Load starter example

Challenge: 0 / 22 cleared

Quiz: order: Before cleaning generated files, first make a? Answer: backup

Answer:

Show hintCheckNextIdle

Quiz: order: Make backup Quiz: safe Quiz: keep Dry-run Clean blocked Full clean VCD safe? Makefile keep?

Quiz: dry-run Backup has rtl Backup has build No vvp after tb remains Quiz: clean-build Stamp format

Count safe Workflow Restore Quiz: logs Safety lock Verify flow

Safe workflow

1. Backup → 2. Dry-run clean → 3. Clean → 4. Verify source remains

Safe to delete

- build/ sim binaries
- logs/ run logs
- *.vcd waveforms

Keep (never wipe)

- rtl/ tb/ docs/
- Makefile README.md
- Anything you cannot regenerate

Project tree

keep = source safe = generated

chip/
chip/Makefile
chip/README.md
chip/rtl/
chip/rtl/top.v
chip/rtl/alu.v
chip/tb/
chip/tb/tb_top.v
chip/docs/
chip/docs/spec.md
chip/build/
chip/build/out.vvp
chip/build/wave.vcd

Actions

Backup: none Cleaned: no Safe left: 6

Backup – timestamped tar.gz

Clean --dry-run (preview only)

Clean for real (needs backup)

Restore tree from starter

Backup artifact

(no backup yet)

Log

Starter loaded – backup – dry-run – clean

Cheat sheet

Step	Example
Backup	tar czf backup_\$(date +%Y%m%d).tar.gz chip/
Dry-run	./clean.sh --dry-run (list only)
Clean	rm -rf build logs *.vcd (after backup)
Verify	ls rtl tb Makefile still present

- Clean build means remove generated artifacts so the next build is fresh – not delete the project.
- If unsure, dry-run first. If still unsure, backup first.

Backup and clean lab

learn_unix — backup clean

Real shell practice

```
wsl - module21-backup-clean/examples/clean_build

# Track A - real Unix
# real shell session (Track A)

$ find sample_project/build sample_project/logs -type f
sample_project/build/output.o
sample_project/build/output.tmp
sample_project/build/wave.vcd
sample_project/logs/run.log

$ tar czf /tmp/m21_backup_demo.tar.gz --exclude=build --exclude=logs -C sample_project .

$ tar tzf /tmp/m21_backup_demo.tar.gz
./
./src/
./src/main.v
./tb/
./tb/test_main.v

$ find sample_project/build sample_project/logs -type f
sample_project/build/output.o
sample_project/build/output.tmp
sample_project/build/wave.vcd
sample_project/logs/run.log

# (still present - clean only after you mean to)
```

Real shell — backup then list clean targets

Real shell practice — try these

```
# find ... -type f — list generated files under build/ and logs/  
find sample_project/build sample_project/logs -type f
```

```
# tar czf backup_...tar.gz — timestamped archive excluding build and  
logs
```

```
tar czf /tmp/m21_backup_demo.tar.gz \  
  
    --exclude='build' --exclude='logs' \  
  
    -C sample_project .
```

```
# tar tzf ... — list what was backed up (no build/logs)  
tar tzf /tmp/m21_backup_demo.tar.gz
```

```
# find again — dry-run view of clean targets (do not delete yet)  
find sample_project/build sample_project/logs -type f
```

Pitfalls to watch

- Never clean before you have a backup you can restore
- Do not run recursive delete on a path you have not listed
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish the backup → dry-run → clean flow
- On the real shell, make a timestamped archive and list clean targets before you delete
- When you are ready, take the short quiz, then continue to relative symlink pitfalls